

CN Assignment 2: Identify a real-world application for both parallel computing and networked systems. Explain how these technologies are used and why they are important in that context.

Real-world Application: online Multiplayer gaming
(eg: Fortnite, Call of Duty)

i) use of parallel computing:

parallel computing is the simultaneous use of multiple processors or cores to solve a computational problem faster.

In online gaming:

- Game physics and Graphics: Game engines use parallel computing to handle complex tasks like physics calculations, lighting, shadows, rendering high-quality 3D graphics and animations. These processes are split across multiple CPU and GPU core.
- AI and Environment Simulation: NPC (non-player) characters' behaviour, environment updates and collision detection are processed in parallel to improve game realism and responsiveness.

- performance Boost : parallel threads run simultaneously, reducing lag and improving frames rates, especially in graphically intense scenes.

importance

- ensures smooth gameplay, fast response time and realistic experience.
- Enables real-time simulation of large environments and multiple players.

(ii) use of Networked system :

A networked system is a collection of interconnected devices that share data and resources, often over the internet.

In online gaming

- Multiplayer connectivity : players from different locations connect to a central server or peer-to-peer network to play in real-time.
- Data Synchronization : player positions, actions, game events and scores are constantly synchronized between devices.
- Game update and cloud storage : Games download patches, updates and store user progress through networked system.

Importance

- Enables real-time interaction between global players.
- Supports massive multiplayer environments
- Reduces delays through optimized server communication and edge computing.